Ian Green

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SUMMARY

Data engineer with 5 years building petabyte-scale data infrastructure in high-security environments. Migrated 1 PiB from file systems to an S3/PostgreSQL lake-house, replaced manual ETL with Airflow pipelines instrumented for lineage and observability, and delivered the organization's first production RAG system. MS Computer Science, Johns Hopkins.

EXPERIENCE

Johns Hopkins Applied Physics Laboratory, Laurel, MD May 2021 - Present

Analysis & Live Test Lead (concurrent) February 2024 - Present

- Built an aggregate stochastic model of EW systems over a unified simulated/live data model, cutting turnaround on emergent analysis requests 5x
- Developed a log data model to extract and normalize failure signals from disparate systems (1TB/hr) and trained a failure-prediction model to predict and flag incidents minutes before impact
- Redesigned analysis team workflow, reducing standard turnaround times from 6 months to 3 weeks
- Accelerated IQ-based radar simulations runtime from 12 to 5 hours through memory layout updates and GPU acceleration of critical code, including custom shaders
- Supervised a $5 million project, including staff mentoring, budget management, overall technical direction, grant proposals, and sponsor interfacing - securing a consistent 8% budget growth

Data Engineering Lead (concurrent) January 2023 - Present

Data Engineer May 2022 - January 2023

Intern May 2021 - May 2022

- Migrated manual ETL process into Airflow, adding data provenance with DataHub and observability with Grafana, reducing ingestion time 75%
- Transferred program data (1 PiB) to S3/PostgreSQL lake-house from file systems, reducing mean access time by 60% and improving security posture to meet IL6 requirements to ship to highly secure environments
- Built a RAG pipeline over a 50TB corpus of documentation for the company's first natural language knowledge hub, now serving 200 engineers over 16 programs
- Automated dependency management with agentic pipelines, reducing mean time to remove packages with critical or high CVEs from 5 days to 3 hours
- Improved pipeline test coverage from 10% to 90% using Pytest and Gitlab CI/CD
- Built a React-based app with microservice backend on OpenShift (Kubernetes) for data visualization and planning using disparate data sources and delivered to highly-classified secure cloud environments

SKILLS

Languages : Python, TypeScript (React.js), HTML/CSS, SQL (PostgreSQL, MySQL), Rust, C++, Java, CUDA
Data Engineering : Apache Airflow, MongoDB, PostgreSQL, DataHub, InfluxDB, S3, Parquet
Infrastructure : OpenShift/Kubernetes, Docker/Podman, slurm, Gitlab CI/CD, JFrog Artifactory
AI/ML : RAG Pipelines, embeddings, vector search, LLM applications, predictive & stochastic modeling, feature engineering
Clearance : Top Secret / Sensitive Compartmented Information (TS/SCI) issued by DoD (Feb. 2026 Investigation)

EDUCATION

Johns Hopkins University August 2022 - May 2025

M.S. in Computer Science

Loyola Marymount University August 2018 - May 2022

B.S. in Computer Science
B.S.E. in Electrical Engineering